# Field observation — concrete structure near the 1948 node

Ladera Ranch vicinity, Orange County, California · recorded 2026-08-07 · LHDRS field record

**THIS IS NOT AN IDENTIFICATION.**
The structure below has not been measured to a standard, has not been sampled, and has not been examined by anyone qualified to identify it. Dimensions are as reported by the observer and are approximate. This document exists to record what was seen, set out what would confirm or exclude a cattle-dipping vat, and be handed to someone able to decide.

**SAFETY — READ BEFORE RETURNING TO THE SITE**
If this *is* a dipping vat, the sediment inside it is the single most concentrated place arsenic would be found anywhere in this landscape — this project's own modelling puts that compartment in the thousands of mg/kg against a California background of 1–11 mg/kg. **Do not dig in it. Do not handle or remove the sediment. Do not let children or animals into it.** Photograph and measure from outside. Arsenic does not degrade; a century has not reduced it.

## The structure as found

Observer photograph, source: vat_photo.jpg. Looking along the axis of the structure. A rusted metal pipe frame lies across the near end; the concrete channel runs away from the camera into heavy dry brush. Depth is obscured by accumulated sediment and vegetation.

## What was observed

| Attribute | As reported | Confidence |
|---|---|---|

| | | |
|---|---|---|
| Material | Concrete | observer, direct |
| Length | approximately 15–20 ft | estimated, not measured |
| Width | approximately 3–4 ft | estimated, not measured |
| Depth | NOT MEASURED — obscured by infill | unknown |
| Form | Discrete linear channel with ends; not a continuous ditch run | observer, direct |
| Associated | Rusted metal pipe frame / rail at one end | observer, direct |
| Setting | Heavy dry brush, drainage corridor, near the 1948 mapped structure | observer, direct |
| Coordinates | NOT RECORDED — no GPS in the image metadata | missing |
| Image | 640 x 480 px, no EXIF GPS or timestamp retained | low resolution |

## The federal specification for a concrete dipping vat

USDA Bureau of Animal Industry Circulars 183 (1911) and 207 (1912) give construction specifications for **concrete** cattle-dipping vats. Circular 174 (1911) states cement is *preferable* to lumber because it "has not the disadvantage of leaking, which is common in wooden vats." **Concrete construction is therefore consistent with, and in fact the recommended form of, a period dipping vat.**

| Dimension | Circular 183 / 207 concrete vat | Structure as reported |
|---|---|---|
| Length | 26 ft at top, tapering to 12 ft at bottom | ~15–20 ft |
| Width | 3.0 ft at top, tapering to 1.5 ft at bottom | ~3–4 ft |
| Depth | 6.5 ft | not measured |
| Capacity | 1,470 gal at 5 ft 3 in fill | unknown |
| Chute | 30 in wide, 20 ft long, leading in | not observed |
| Drip pen | ~36 in wide, 20–40 ft, sloped back toward vat | not observed |

## Why it could be a vat — the case for

- **Concrete is the specified material.** Not merely compatible — Circular 174 recommends it over timber specifically because timber vats leaked.
- **Reported length falls inside the spec envelope.** The documented vat is 12 ft at the floor widening to 26 ft at the rim. A reported 15–20 ft is what an observer would measure on a *tapered* vat part-filled with sediment, reading the length at whatever level the infill now sits — not at the true floor or rim.
- **Reported width brackets the spec top width.** 3–4 ft observed against 3.0 ft specified.
- **It is a discrete structure with ends.** A concrete-lined drainage channel runs continuously across a slope; it does not begin and end in 15–20 ft. A vat does.
- **A metal frame is present.** Vats were fitted with rails, splash boards on hinged posts, and chute fencing. A rusted pipe frame at one end is consistent with that, though not diagnostic.

- **The setting fits the siting logic.** Vats were placed near water and stock-handling ground. The structure sits in a drainage corridor near the only mapped 1948 structure in the study area.

## Why it might not be — the case against

Stated at equal length, because a document that only argues one way is worth nothing to the agency or specialist who has to act on it.

- **Depth is unmeasured, and depth is the whole question.** A dipping vat is **6.5 ft deep** with steep sides — cattle swim through it. If this structure proves to be 1–2 ft deep it is a trough, a flume or a lined ditch, and nothing else about it matters.
- **Concrete-lined drainage is extremely common here.** Brow ditches, V-ditches and terrace drains with pipe railings are standard on California graded slopes and in flood-control corridors. Many are 3–4 ft wide.
- **No taper has been confirmed.** A vat narrows markedly from rim to floor. A ditch has parallel walls. This has not been checked.
- **Neither end feature has been observed.** A vat is asymmetric — a steep entry slide at roughly 25° at one end, a cleated exit ramp at the other. Their absence would be strong evidence against.
- **No drip pen or chute has been seen.** These are large, concrete, and adjacent. Their absence is not conclusive — they may be buried or demolished — but their presence would be near decisive.
- **The systematic aerial search found no vat in this area.** 37,228 cells across the 1929, 1938 and 1947 frames returned nothing vat-like inside the study frame. That is weak evidence of absence, but it is on the record and must be weighed.
- **Age is unestablished.** Nothing observed dates the concrete. It could postdate the dipping era by decades.

## What would resolve it — in priority order

| # | Measurement | Why it decides | Vat if… |
|---|---|---|---|
| 1 | Depth to hard concrete floor, probed through sediment at several points | The single most discriminating number. Vats are deep because cattle swim. | 5–6.5 ft |
| 2 | Width at rim and again as low as reachable | A vat tapers; a ditch has parallel walls. Requires no digging. | narrows ~3 ft → ~1.5 ft |
| 3 | Photograph both ends separately | A vat is asymmetric: steep slide in, cleated ramp out. A ditch is uniform. | slide one end, ramp the other |
| 4 | Look for an adjacent flat concrete apron | The drip pen slopes back toward the vat and is unmistakable once seen. | apron 36 in × 20–40 ft |
| 5 | Total length rim to rim, plus GPS coordinates and a scale object in frame | Makes the record usable by anyone else and locatable again. | 12–26 ft |

### Capacity check, once depth is known

Using the reported 15–20 ft × 3–4 ft footprint and the prismatoid volume the circulars themselves specify: a depth of 2 ft yields about 916 gal; 3 ft about 1,375 gal; 4 ft about 1,833 gal; 5.25 ft about 2,405 gal; 6.5 ft about 2,978 gal. The documented capacities are **1,470 gal** (Circular 183/207 concrete vat) and **2,088 gal** (Circular 174 swim vat). Depths at or above roughly 4 ft put the structure in the documented range; depths under 3 ft place it outside.

## If it holds up

A suspected historic arsenical site is a regulatory matter with an established pathway. It should not be handled privately or by a contractor.

- **California DTSC** — the agency for suspected historic contamination. EnviroStor is their public system and shows no record for this area, which is itself a documented gap.
- **OC Public Works / Flood Control** — if the structure sits in or beside a county drainage easement, they hold the as-builts and the maintenance history.
- **Property owner consent** — LARMAC, the County, or Rancho Mission Viejo depending on parcel. This project's gate G10 already anticipates exactly this and requires consent plus an accredited laboratory.
- **Analyte suite:** arsenic *and lead*, plus DDT/DDE, toxaphene and PAHs. The lead ratio is what separates cattle-dip ground from orchard ground — dip ground carries arsenic without lead.

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**Statement class:** field observation, unverified. **Provenance:** observer photograph and verbal dimensions, uncorroborated. Specifications quoted are A1 — USDA Bureau of Animal Industry Circulars 174 (1911), 183 (1911) and 207 (1912), held locally with full text and SHA-256 checksums.
**This document does not assert that a dipping vat has been found, that arsenic is present at this or any location, that anyone has been exposed, or that any illness has any environmental cause.** It records an observation and the measurements that would settle what it is. A single accredited laboratory result would outweigh everything in it.
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